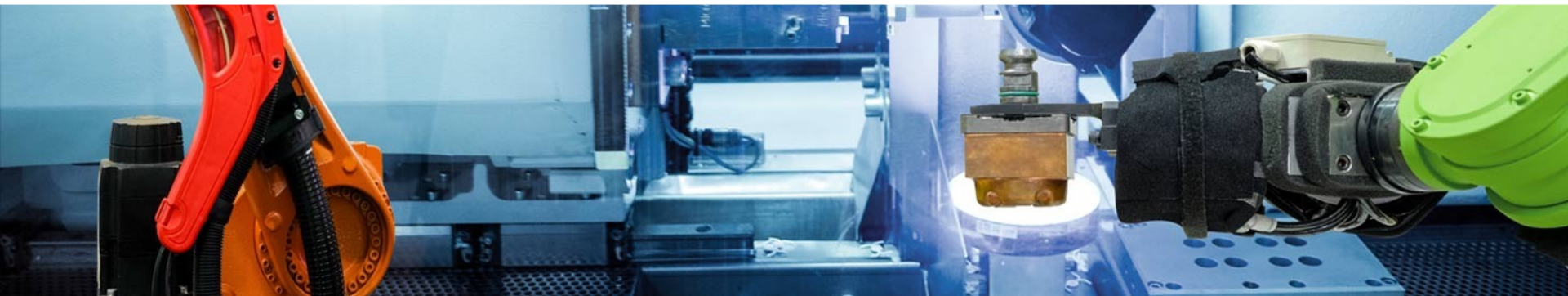




Mechatronic system design techniques

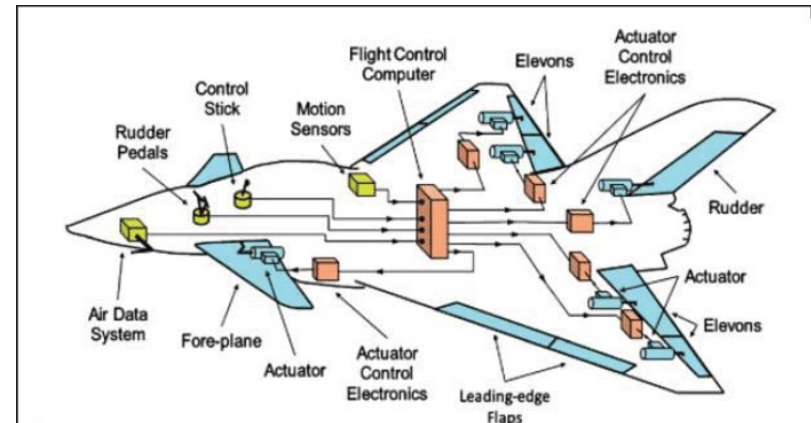
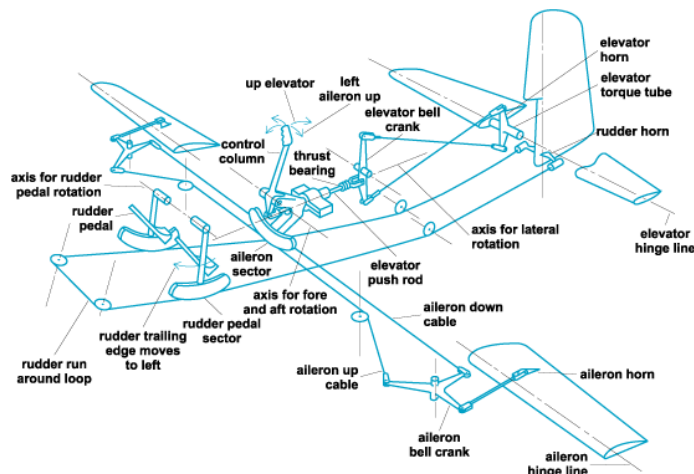


Specific Requirements to the Design of Mechatronic Systems

Requirements can be grouped into two categories: **1) Impacts of complexity** and **2) impacts of heterogeneity**.

□ Impacts of Complexity

- Mechatronic systems are characterised by high complexity due to the cross-linkage of different engineering domains.
- The complexity is caused by the increasing number of connected elements. Mechatronics does not only improve behavior, precision or the costvalue ratio of known solution principles.
- It especially creates new functions: Active suspension and electronic stability features in the automotive industry or fly-by-wire in aeronautics



Specific Requirements to the Design of Mechatronic Systems

□ Impacts of Complexity

- 1. Procedure with changing level of detail and abstraction:** During the design process the designing engineers need the reciprocal action between the overall context and the detail focus. Because of the cross-linkage to other elements new detail solutions and findings have to be checked permanently in the overall context as well.
- 2. Methods of structuring and hierarchisation:** Mechatronic systems should be structured in order to reduce complexity and to minimise interactions, e.g. the hierarchical structure in basic elements, systems and linked systems.
- 3. Early modelling and simulation:** Interactions between sub-systems, e.g. coupling of engine management and brake system for the electronic stability program (ESP), can only be anticipated by the early modelling and simulation of the system behavior. Suited methods and software tools are necessary.
- 4. Integration and verification/validation of properties:** The design results have to be integrated to an overall system and to be checked continuously by means of the specified solution concept and the requirements. It is to be assured that the actual system characteristics match with those wanted. Hardware-in-the-Loop (HIL) plays an important role to analyse real and virtual components in a common environment.

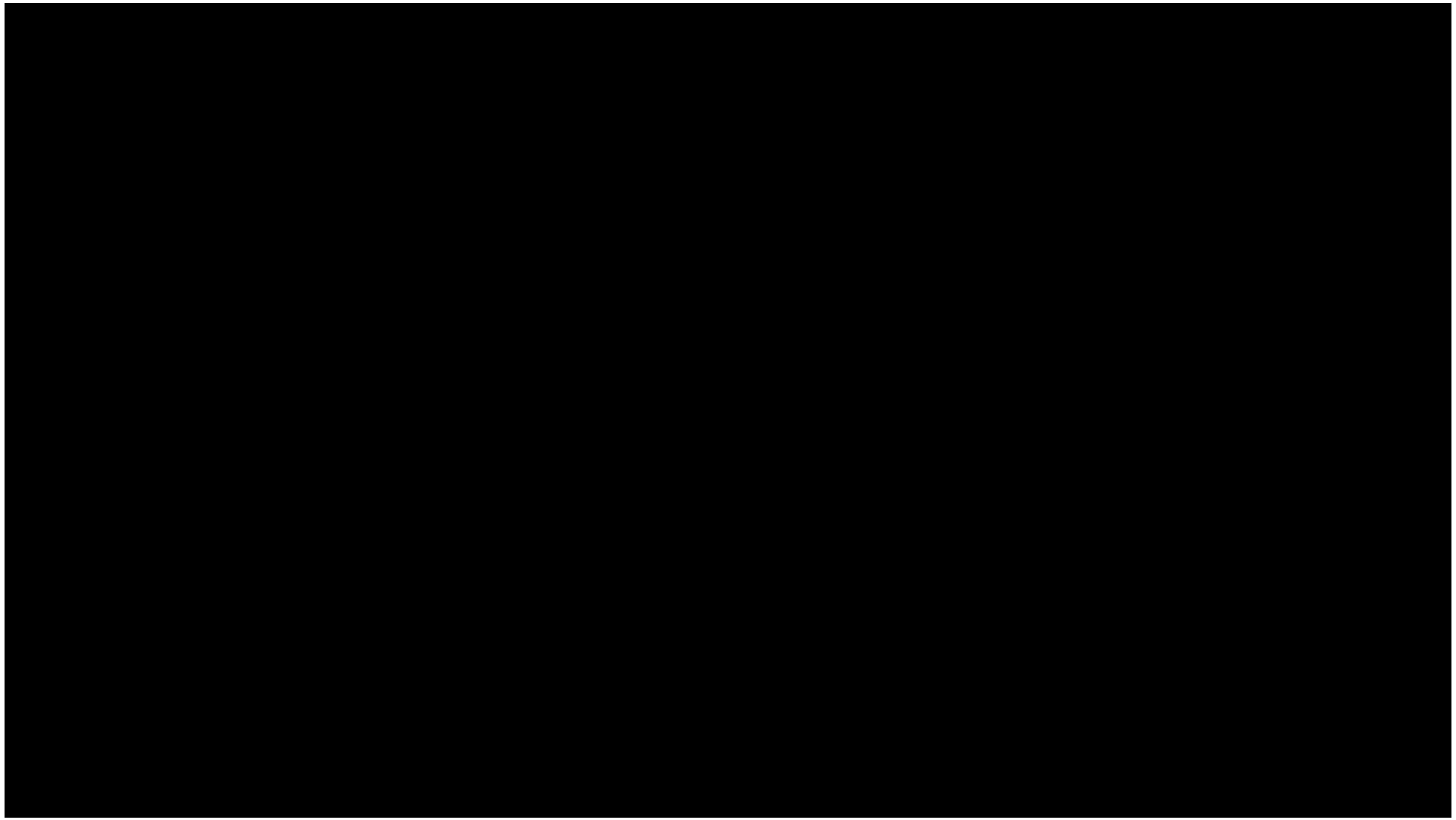
Specific Requirements to the Design of Mechatronic Systems

❏ Impacts of Heterogeneity

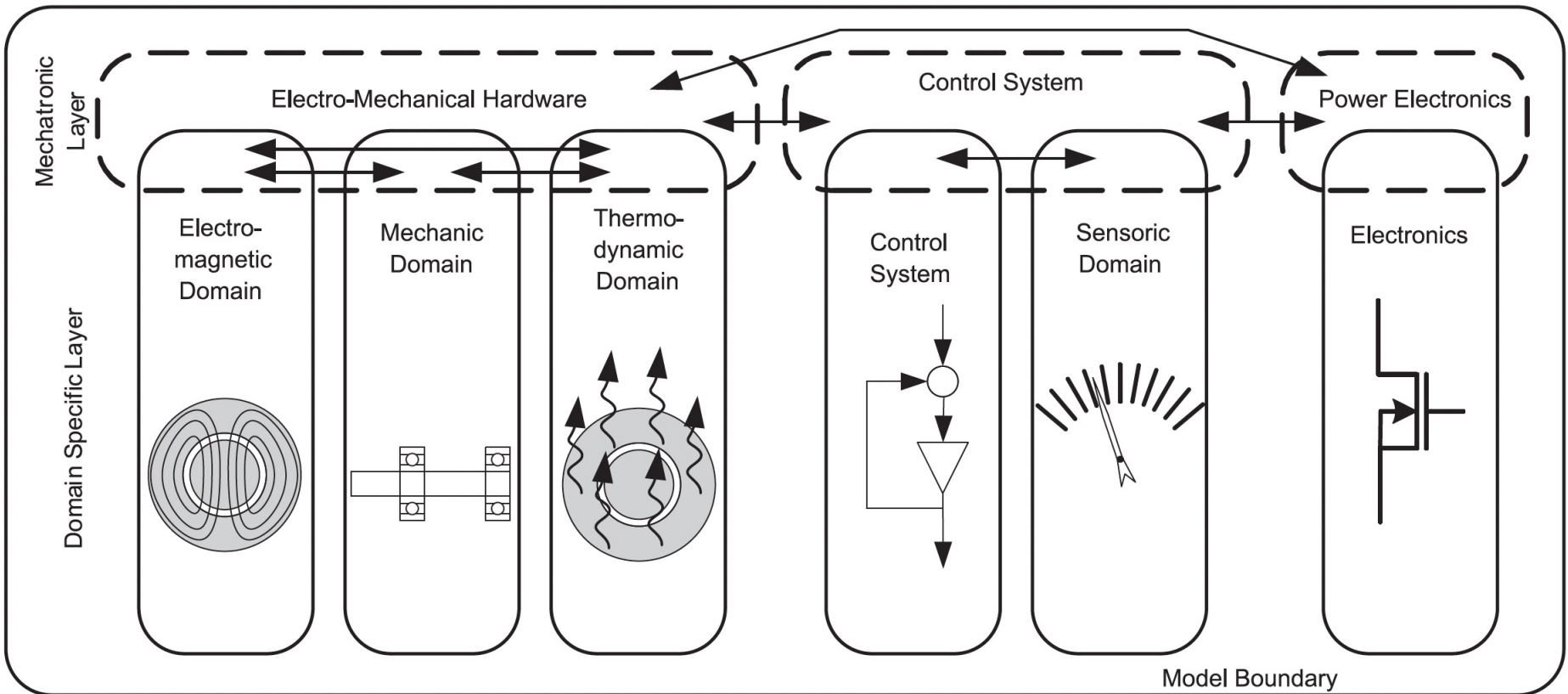
Solutions and components are coming from different engineering domains such as mechanics, electronics and information technology. These involved domains are working on the basis of established, specific design methods, their own ways of thinking, nomenclatures and experiences. The integration of heterogeneous components to mechatronic systems is a big challenge expressed by four requirements:

- 1. Cross-domain collaboration in teams:** Communication and co-operation between the involved disciplines is required to bring together synergetic knowledge and to design an entirely optimised solution.
- 2. Cross-domain specification:** Experts from different disciplines need a common method to specify the results of product conceptualisation in a cross-domain way [9].
- 3. Partitioning:** The partitioning, i.e. the distribution into working principles/solution elements of the involved domains and their assignment to functions, is an important step. Especially the heterogeneous partitioning needs methodical support.
- 4. Exchange/integration of models:** The design results – modelled by the domains with different model languages and tools – need to be integrated. Procedures e.g. MechaSTEP and new languages e.g. Modelica are necessary.

Challenges in Design



Challenges in Design

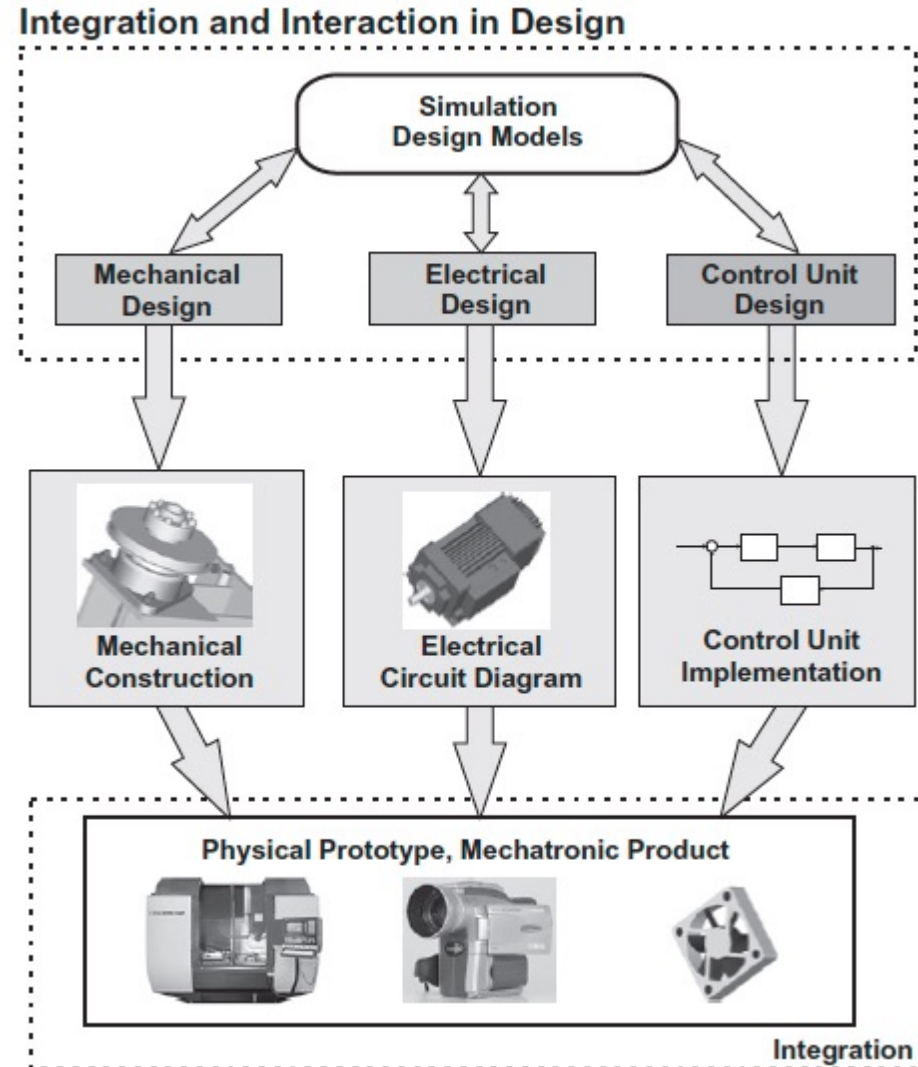


Mechatronic coupling, define mechatronic module of the PMSM drive system

Challenges in Design

1. Integration of Heterogeneous Subsystems

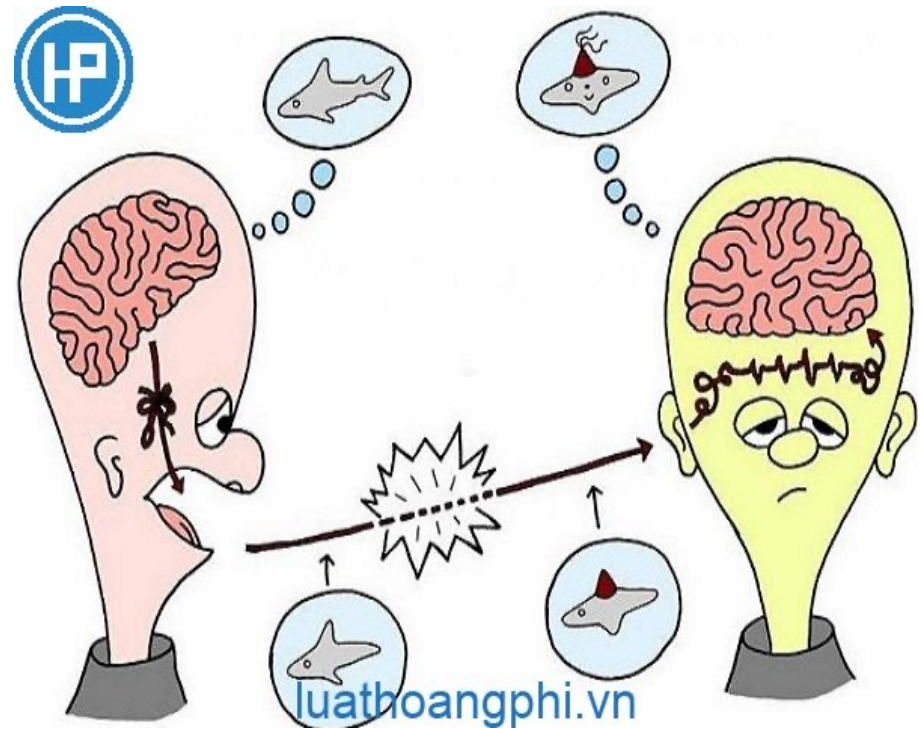
- **Challenge:** Mechatronic systems often integrate subsystems from multiple engineering domains (mechanical, electrical, software). Ensuring compatibility and smooth interaction among these domains can be difficult.
- **Solution:** Use standardized modeling languages such as Modelica and SysML to facilitate integration. These tools allow for cross-domain collaboration and provide a unified platform for designing mechatronic systems.



Challenges in Design

2. Communication with Customers and among Design Team Members Is Challenging

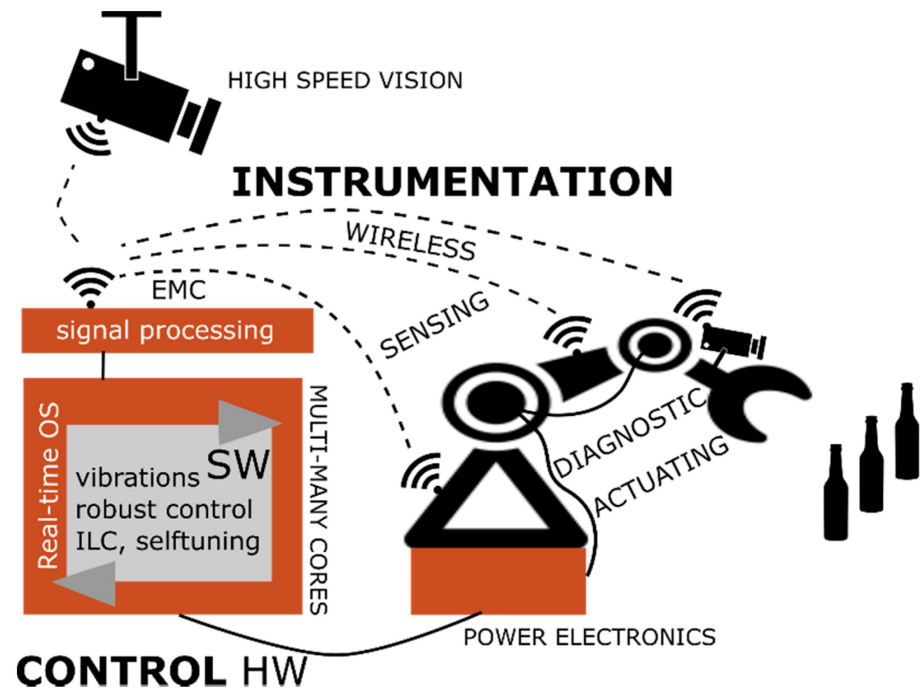
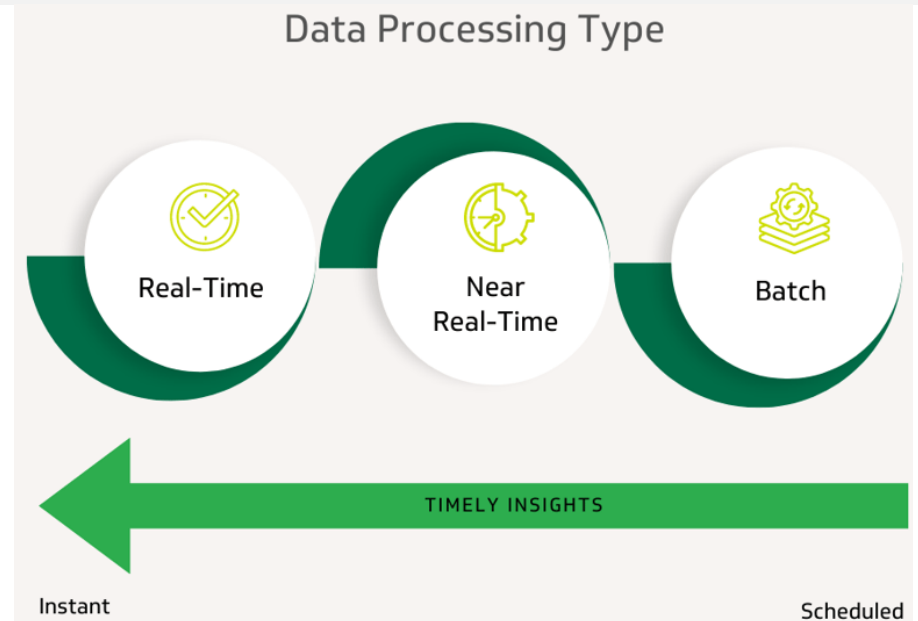
- **Challenge:** Understanding customer requirements and conveying design plans can be a lengthy and complex process, potentially leading to misunderstandings and unsuitable designs.
- **Solution:** Implement a mechatronics design approach that integrates mechanical, electrical, and control design tools. Create a virtual machine prototype before physical design to test and refine the machine's design in software.



Challenges in Design

3. Real-time control

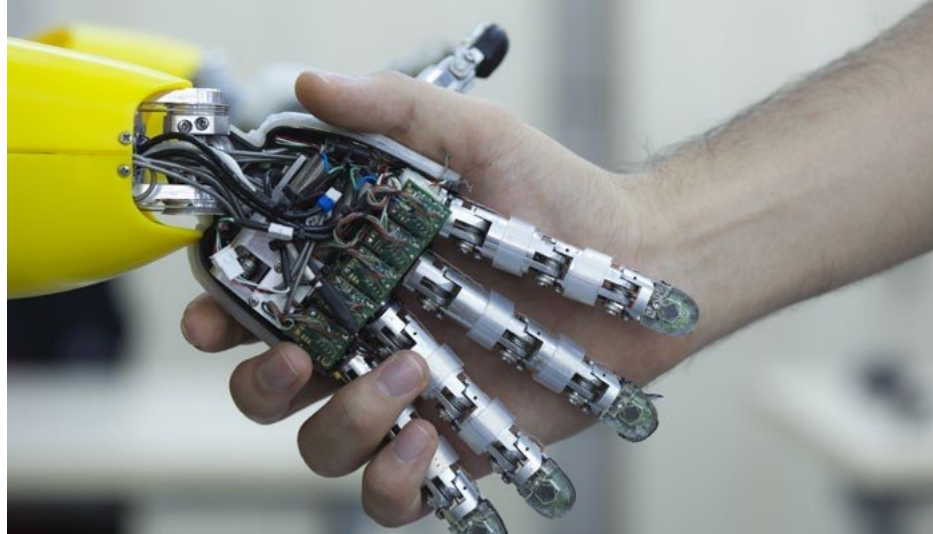
- **Challenge:** Mechatronic systems require real-time control and data processing, especially in environments where precision is critical. Managing large volumes of sensor data and ensuring timely responses is a major challenge.
- **Solution:** Implement real-time operating systems (RTOS) and distributed control systems. These systems ensure low-latency data processing and precise execution of control tasks.



Challenges in Design

4. High reliability and safety :

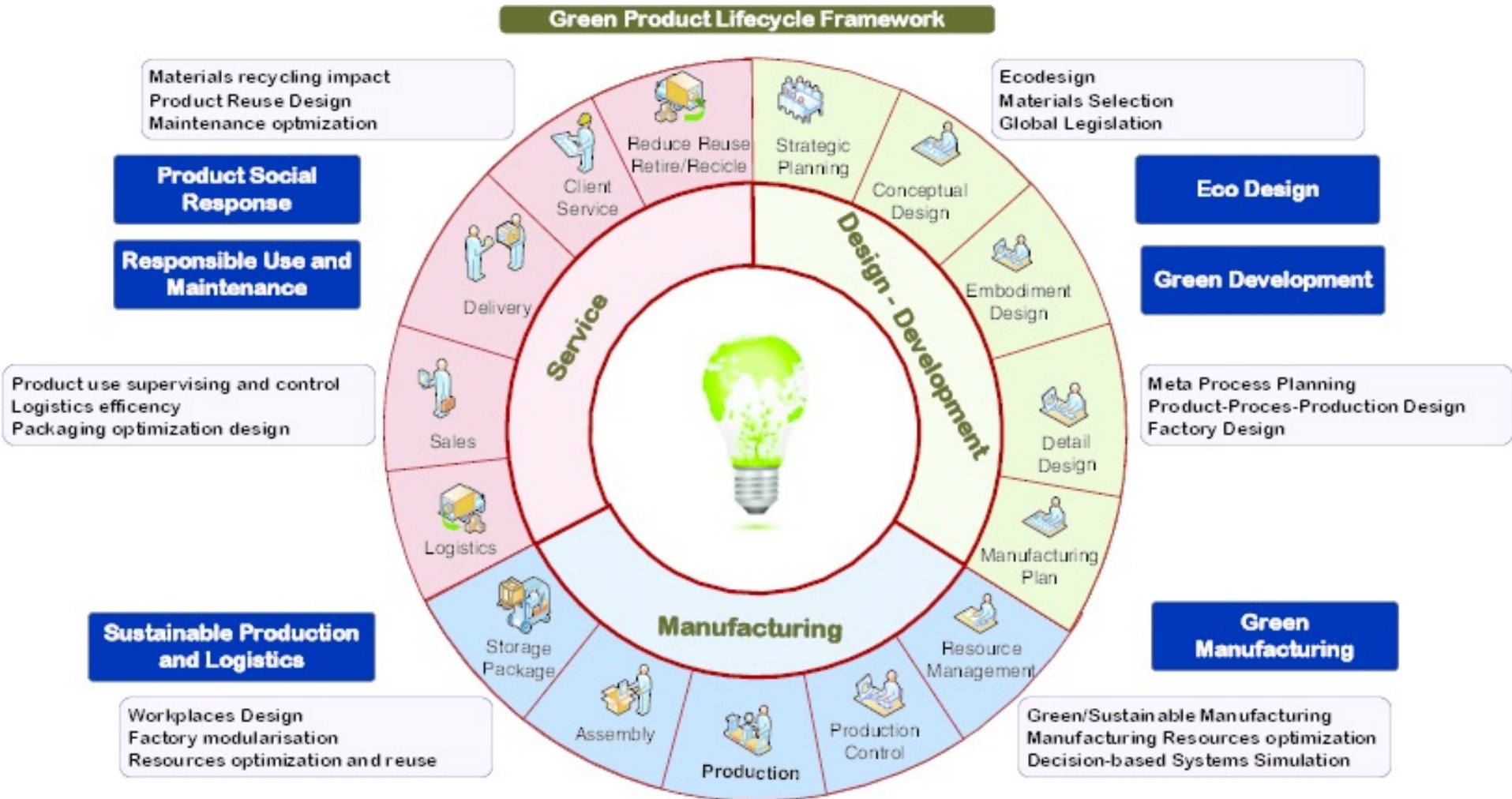
- **Challenge:** High reliability and safety are essential in mechatronic systems, especially in critical applications such as autonomous vehicles or medical devices. System failures can lead to significant risks.
- **Solution:** Use redundancy in critical systems and apply fault detection and isolation (FDI) algorithms. Conduct thorough Hardware-in-the-Loop (HIL) testing to ensure system safety under real-world conditions.





Design techniques

1. Life cycle design



- **Definition:** A methodology that considers the entire life cycle of a product from inception to end-of-life, focusing on reducing environmental impacts and improving sustainability.

1. Life cycle design

Key Phases:

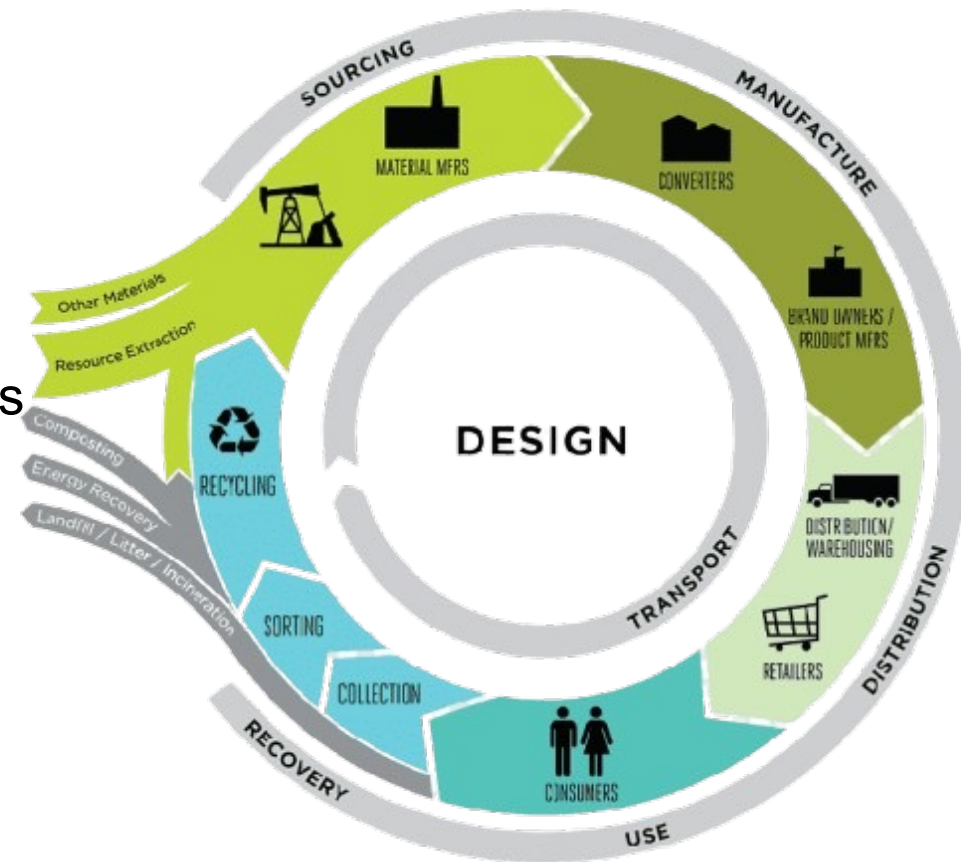
1.Sourcing: Involves extracting raw materials and resources needed for production.

2.Manufacturing: Converting raw materials into usable products through various processes like assembly and fabrication.

3.Distribution: Products are transported from manufacturing facilities to retailers and warehouses, ready for distribution to consumers.

4.Use: Consumers utilize the products. In mechatronic systems, this includes everything from robotic systems to electric vehicles in day-to-day operations.

5.Recovery: After the product's useful life, the materials are collected, sorted, and processed for recycling or disposal.



1. Life cycle design

Principles and Strategies:

- **Environmental Impact Reduction:** It aims to minimize negative environmental effects by reducing energy consumption, emissions, and resource depletion.
- **Sustainability:** It considers sustainability principles, such as reducing waste, promoting recycling, and using renewable resources when possible.
- **Cost Efficiency:** It strives to reduce life-cycle costs by designing for longevity, ease of maintenance, and efficient use of resources.
- **Responsibility:** It encourages responsible resource management and adherence to environmental regulations.



1. Life cycle design

Design Considerations:

- **Material Selection:** It focuses on selecting materials that are environmentally friendly, durable, and can be recycled or disposed of responsibly.
- **Energy Efficiency:** It emphasizes designing for energy efficiency during both operation and manufacturing.
- **Durability and Reliability:** It promotes the design of products that have longer lifespans and require fewer repairs or replacements.
- **Ease of Disassembly:** It considers the ease of disassembling products at the end of their life to facilitate recycling or responsible disposal.



1. Life cycle design

Pros:

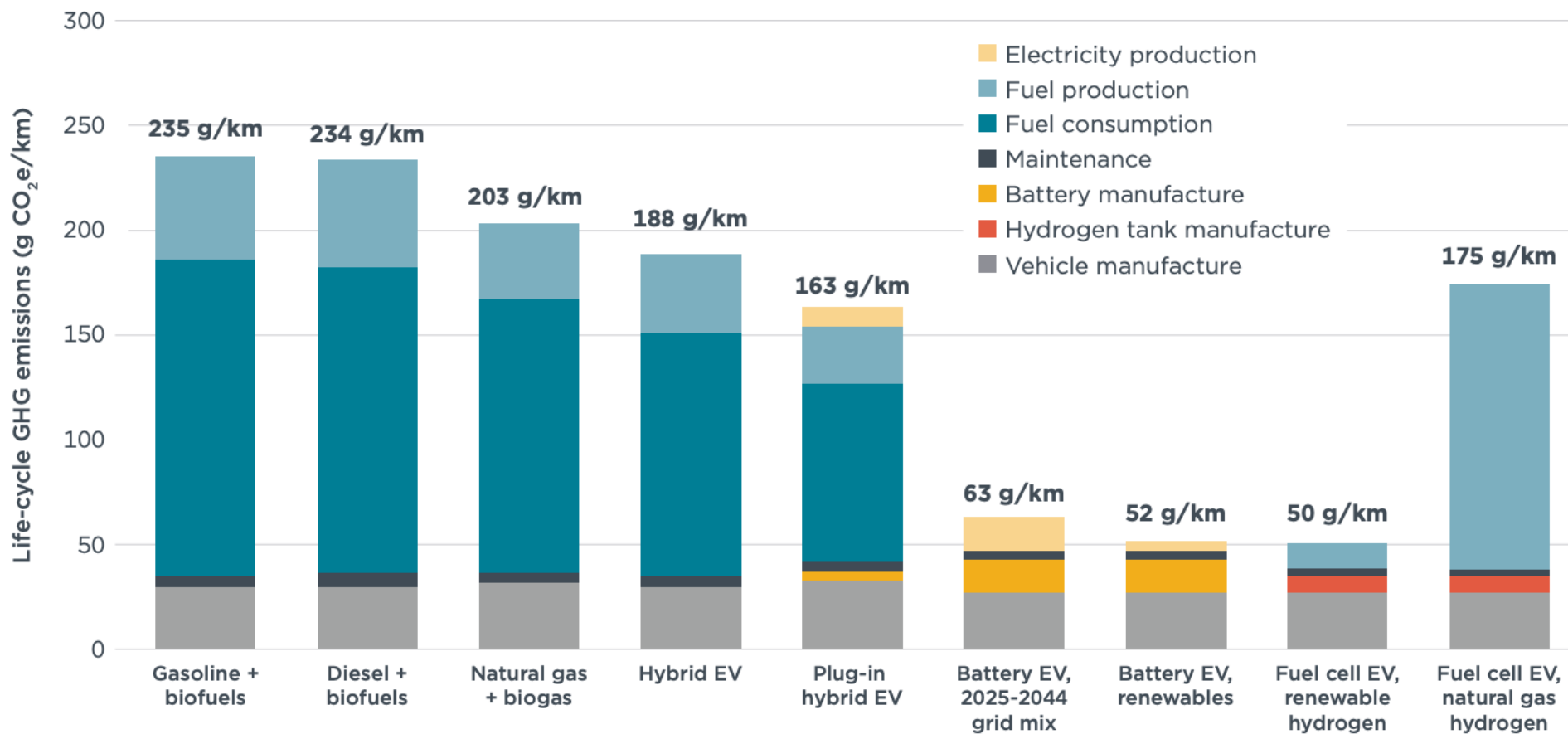
- **Eco-Friendly:** It helps make products and systems more environmentally friendly by considering their impact from start to finish.
- **Cost Savings:** It can save money in the long run by reducing waste, energy use, and resource consumption.
- **Reduced Risk:** It lowers the risk of environmental or regulatory problems down the line.
- **Innovation:** It encourages creative solutions and can make businesses more competitive.
- **Good Reputation:** It can enhance a company's image by showing commitment to sustainability.

Cons:

- **Complex:** It can be complicated to implement, especially for big projects.
- **Initial Costs:** It might require spending more money upfront for research and development.
- **Balancing Act:** Sometimes, what's environmentally friendly may be more expensive.
- **Data Needed:** It relies on having a lot of data, which might not always be available.
- **Slower Development:** It can slow down the product development process due to extra planning and analysis.

1. Life cycle design

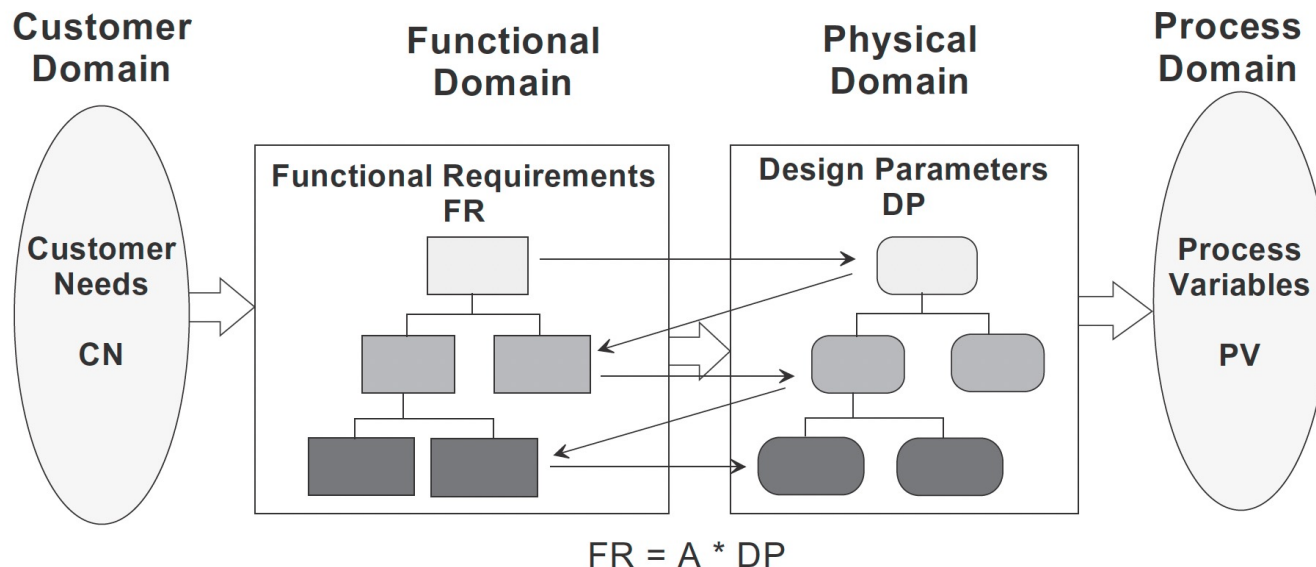
Life-cycle GHG emissions of medium segment passenger cars sold in the European Union in 2025



LIFE-CYCLE GREENHOUSE GAS EMISSIONS FROM PASSENGER CARS IN THE EUROPEAN UNION: A 2025 UPDATE AND KEY FACTORS TO CONSIDER

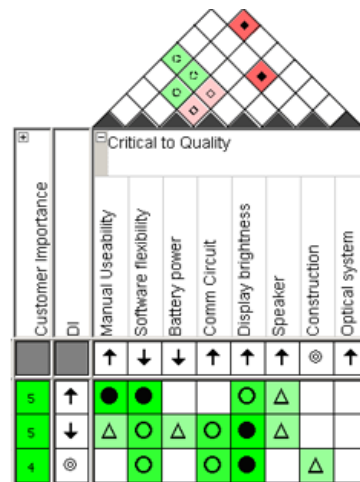
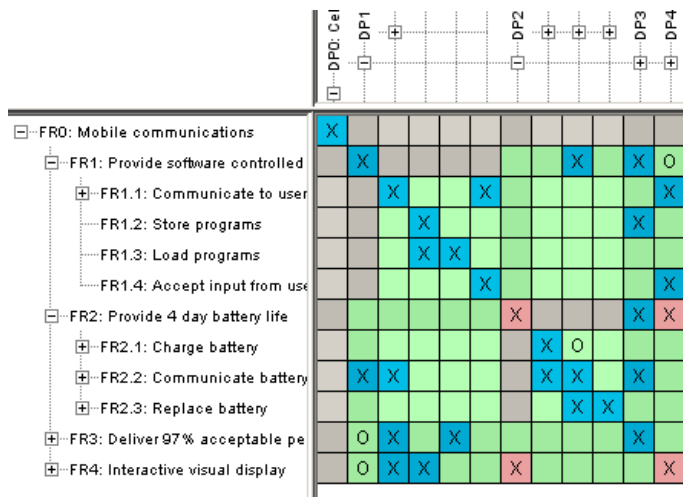
2. Axiomatic design

- **Axiomatic design** is a systems design methodology using matrix methods to **systematically analyze the transformation of customer needs into functional requirements, design parameters, and process variables**.
- Axiomatic design is particularly useful in industries where precision and reliability are critical, such as aerospace, automotive, and manufacturing



2. Axiomatic design

- The customer domain indicating the needs of the customer.
- The functional domain expressing the desired functions (desired behaviour) of the design object.
- The physical domain representing the physical properties of the design object.
- The process domain illustrating how to achieve or produce the design object.



$$\begin{Bmatrix} FR_1 \\ M \\ FR_n \end{Bmatrix} = [A] \begin{Bmatrix} DP_1 \\ M \\ DP_n \end{Bmatrix} \quad (1)$$

	DP 1	DP 2	DP 3
FR 1	X	0	0
FR 2	0	X	0
FR 3	0	0	X

Uncoupled

	DP 1	DP 2	DP 3
FR 1	X	0	0
FR 2	X	X	0
FR 3	X	X	X

Decoupled

	DP 1	DP 2	DP 3
FR 1	X	X	X
FR 2	X	X	X
FR 3	X	X	X

Coupled

2. Axiomatic design

Axiom 1 - The Independence Axiom

- This states that functional requirements (FRs) should be independent of one another.
- In other words, the design should be modular so that changes to one FR do not affect other FRs.
- This enables designs that are less prone to unintended consequences and side effects when modifications are made.
- On a design matrix, an independent design is represented by diagonal or triangular matrix with no couplings between FRs and design parameters (DPs).

Axiom 2 - The Information Axiom

- This states that designs should contain the minimum amount of information required to satisfy the FRs.
- The information content should be minimized to reduce complexity and uncertainty.
- This enables simpler, more robust designs by avoiding unnecessary "information" that does not contribute to meeting the functional goals.
- On a design matrix, the information axiom is satisfied when each FR is addressed by the minimum number of DPs.
- Together, the two axioms promote decoupled, modular design with simplified solutions focused on the essential FRs to be met.

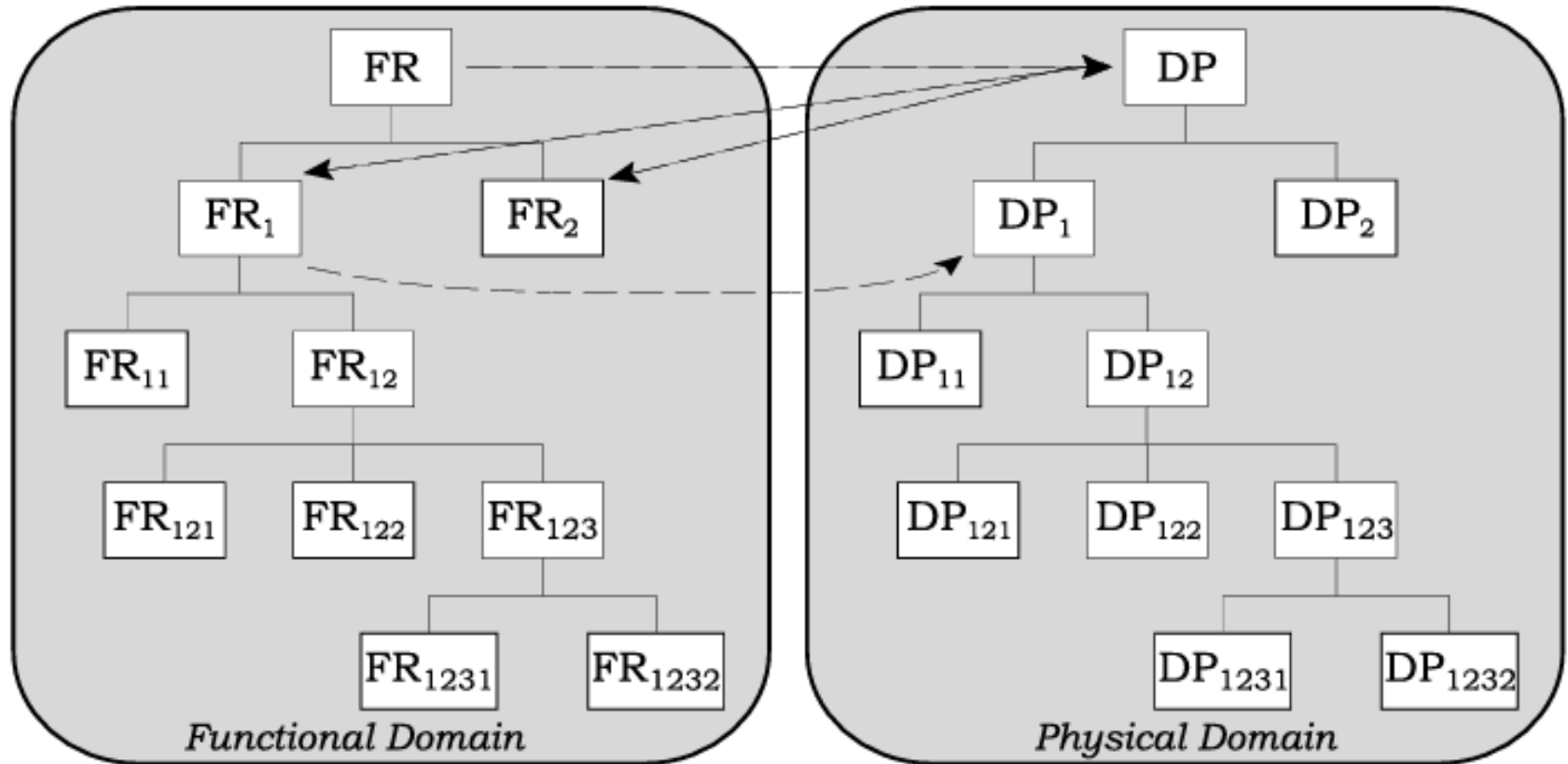
2. Axiomatic design

- The Information Axiom helps to find the best design solution among different possibilities I_i
- The smaller the information content, the less information is needed to reach the design goals
- The information content can be computed by calculating the **probability P_i** of satisfying FR_{*i*}, also known as the **probability of success**

$$I_i = \log_2 \frac{1}{P_i}$$

2. Axiomatic design

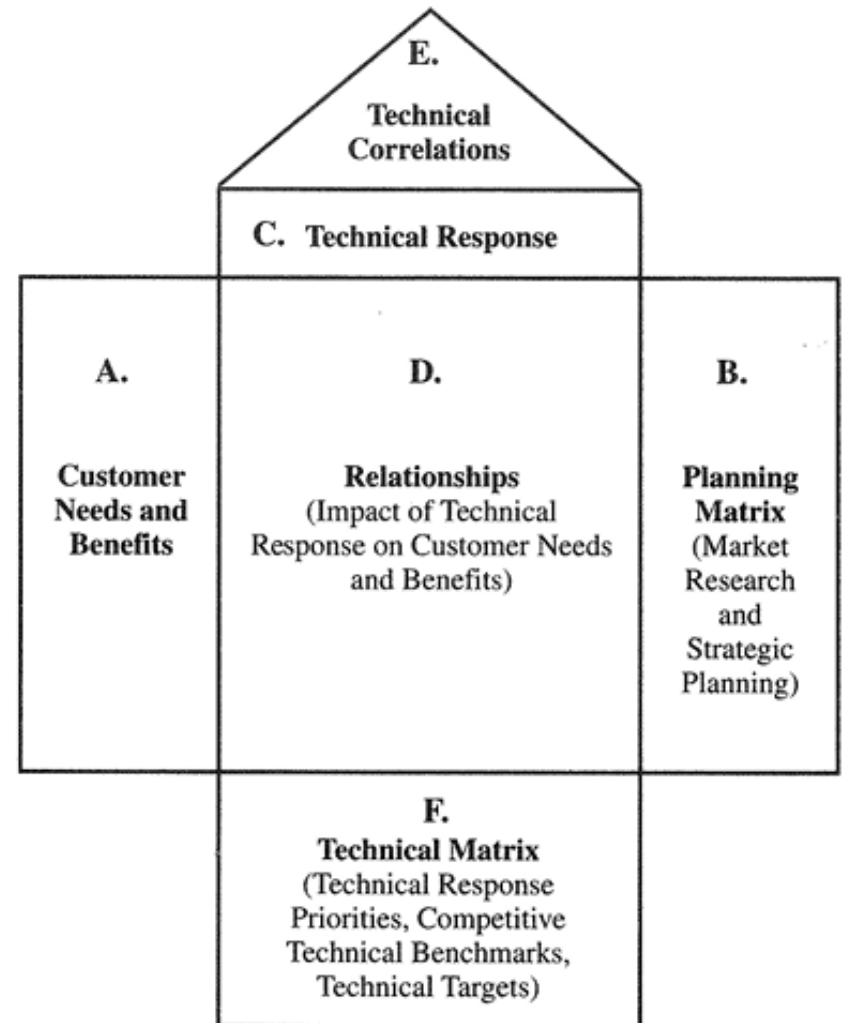
Decomposition, hierarchy and zigzagging



2. Axiomatic design

Quality Function Deployment

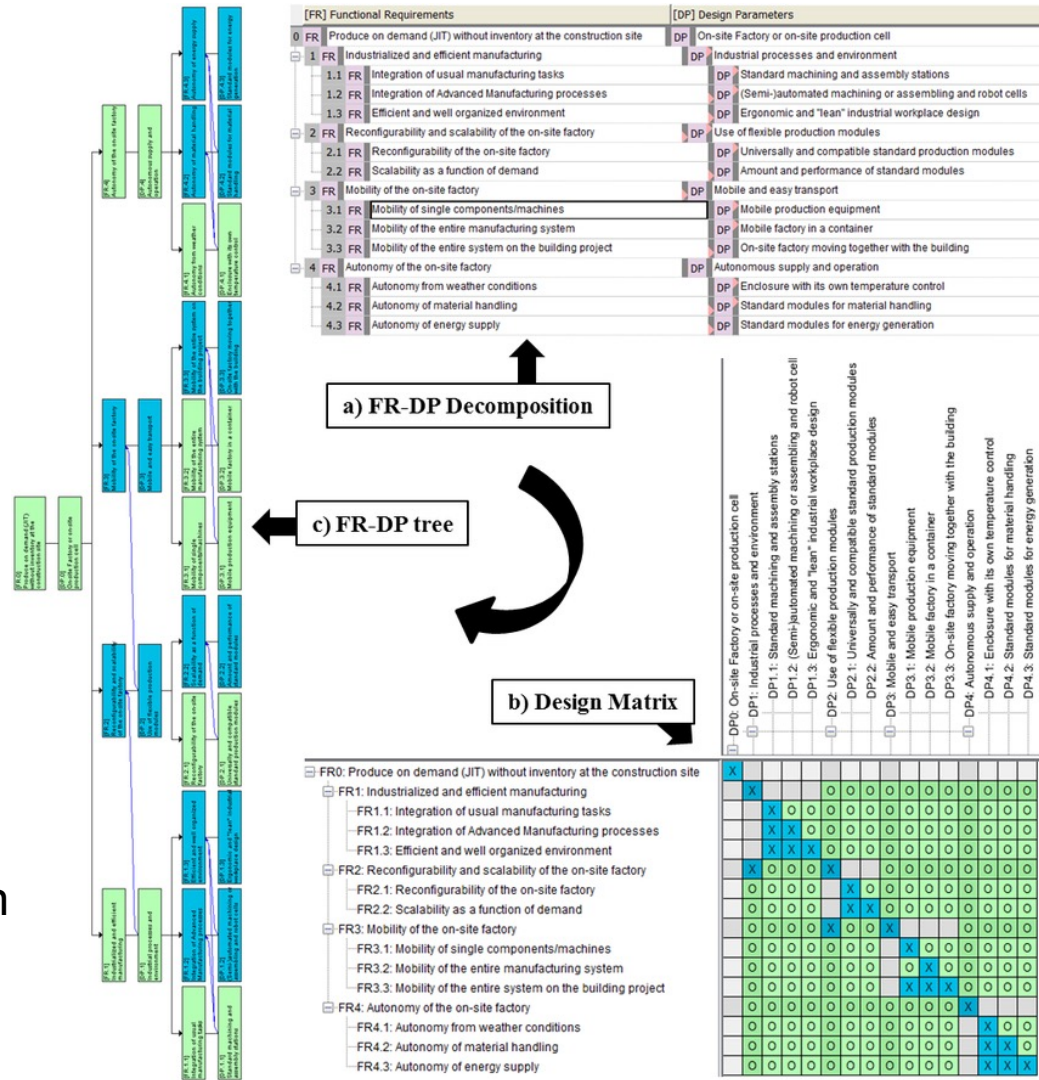
- **Quality Function Deployment (QFD)** is a method for structured product planning and development that follows a process to correlate customer requirements with product properties and the technical specification of the product using a morphological chart.
- **House of Quality**, which consists of a customer needs as well as a technical response section, a relationships matrix, a section for correlations in between technical responses, a planning matrix and a technical matrix



2. Axiomatic design

Key Steps in Axiomatic Design:

- **Identify Customer Needs:** Understand and document customer's requirements.
- **Identify Design Parameters (DPs):** Identify adjustable factors.
- **Create the Domain (D) Matrix:** Quantify relationships between DPs and requirements.
- **Analyze Independence:** Minimize dependencies and ensure that each FR has a unique set of DPs associated with it.
- **Design Matrix (P) and Standardization:** Map DPs to design solutions.
- **Evaluate and Refine:** Assess and optimize the design.



2. Axiomatic design

				a		b		c		d		e		Module information		
				1		2		3		4		5				
				1		2		3		4		5				
				1		2		3		4		5				
Object A	1	Describe the FRs	X											M114121		
	2	Display to the graphics	X	X										M114122		
	3	1	Support copying	X	X	X								M11412341		
		2	Support cutting	X	X		X							M11412342		
		3	Support pasting	X	X			X						M11412343		
		4	Support moving	X	X		X	X						M11412344		
	4	1	Make a new storage	X	X	X	X	X	X					M1141235		
		2	Support changing	X	X	X	X	X	X		X			M1141237		
		3	Support deleting	X	X	X	X	X	X					M1141238		
	Object B	1	Describe the DPs												M114141	
2		Connect the child level												M114142		
3		Display to the graphics	X											M114143		
4		1	Support copying	X	X	X	X	X						M1141441		
		2	Support cutting	X	X	X	X	X						M1141442		
		3	Support pasting	X	X	X	X	X						M1141443		
		4	Support moving	X	X	X	X	X						M1141444		
Object C		1	Make a new storage	X	X	X	X	X	X						M1141445	
		2	Support changing	X	X	X	X	X	X		X				M1141446	
		3	Support deleting	X	X	X	X	X	X						M1141447	
	1	Define the design matrix	X			X	X	X		X	X			M114161		
	2	Display to the graphics	X							X				M114162		
	3	1	Make a new storage	X						X	X				M1141631	
		2	Change each element							X					M11416321	
		3	Change between def DP and Alt DP							X					M11416322	
		4	Change column element							X					M11416323	
	Object D	3	Support copying	X						X	X				M1141633	
4		Check the status of design matrix									X			M1141634		
1		Define the rearranged design matrix	X											M1141621		
2		1	Make a new storage								X	X			M11416221	
		2	Support matrix rearrange								X	X			M11416222	
Object E		1	Define the system architecture	X						X	X				M1141631	
		2	Display to the graphics	X						X					M1141632	
		3	1	Make a new storage	X							X	X	X		M11416331
			2	Find parallel flow								X	X	X		M11416332
		3	2	Find serial flow								X	X	X		M11416333
	3		Find serial flow								X	X	X		M11416334	

2. Axiomatic design

Pros	Cons
Systematic and Structured: Guides designers step-by-step.	Initial Complexity: Time-consuming setup and analysis.
Ensures Independence: Reduces risk of unintended consequences between subsystems.	Hard to Achieve Independence: Some coupling may be inevitable.
Manages Complexity: Reduces complexity by ensuring minimal interaction between parts.	Subjective: Interpretation of FRs and DPs can vary.
Robust and Reliable Designs: Increases system robustness by decoupling functions.	Learning Curve: Requires training to implement correctly.
Encourages Innovation: Prompts exploration of new solutions and opportunities.	Limited Flexibility for Iterative Design: Not ideal for fast prototyping.
Easier Maintenance: Modular design allows easy updates without affecting the whole system.	Limited Tool Support: Fewer integrated tools for some industries.
Quantifiable Design: Information Axiom helps minimize unnecessary complexity.	Challenges in Multidisciplinary Projects: Harder to ensure independence across domains.

2. Axiomatic design

Example: Design a car's suspension system

Step 1: Identify Customer Needs (Functional Requirements):

- Requirement 1 (FR1): Ensure a smooth ride for passengers.
- Requirement 2 (FR2): Provide stability during cornering.
- Requirement 3 (FR3): Absorb shocks from uneven road surfaces.

Step 2: Identify Design Parameters (DPs):

- Parameter 1 (DP1): Spring stiffness.
- Parameter 2 (DP2): Damper (shock absorber) design.
- Parameter 3 (DP3): Tire selection.

2. Axiomatic design

Step 3: Create the Domain (D) Matrix:

	DP1	DP2	DP3
FR1	X	X	X
FR2	X	X	X
FR3	X	X	X

Step 4: Analyze Independence:

Independence Analysis: The initial matrix is coupled, violating the Independence Axiom, as multiple FRs are influenced by the same DPs (e.g., DP2 affects all FRs). Rearranging FR and DP orders cannot achieve a strict diagonal form due to overlapping supports.

2. Axiomatic design

Step 5: Resolve Design Challenges

Apply hierarchical decomposition (zig-zagging) for better independence:

Level 0:

FR0: Provide a safe and comfortable suspension system.

DP0: Integrated suspension system (springs, dampers, tires).

Level 1 Decomposition:

FR1': Control vertical oscillations (vertical ride comfort).

FR2': Control lateral roll (roll stability).

FR3': Absorb local shocks.

Corresponding DPs (refined for independence):

DP1': **Suspension design (springs, dampers).**

DP2': **Anti-roll bar stiffness.**

DP3': Tire selection.

2. Axiomatic design

Step 6: Evaluate and Refine the Design:

	DP1	DP2	DP3
FR1	X		X
FR2	X	X	
FR3			X



	DP3	DP1	DP2
FR3	X		
FR1	X	X	
FR2		X	X

2. Axiomatic design

Case study - ELECTRIC BICYCLES

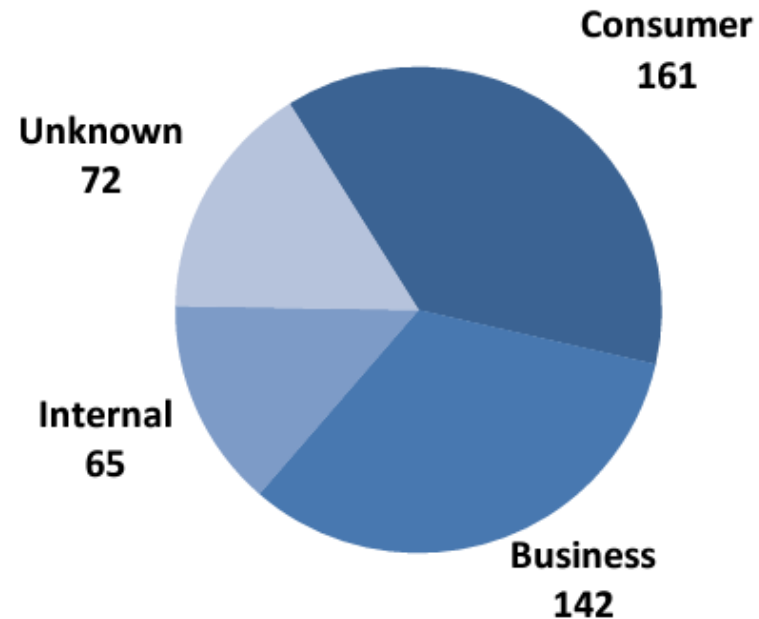


2. Axiomatic design

Step 1: Data collection

- The Axiomatic Design process starts with the customer domain which assesses the needs of customers.
- A total of 440 customer statements were identified
- Sorting and processing of the data was done manually

Customer Statements according to Customer Type



Id.	Customer statement
49	"Quite a lot of force needs to be applied to the brake levers to generate an acceptable deceleration."
62	"The brakes were still sufficient, although you could feel that they were stressed."
217	"Insufficient brakes."

FR₇: Have sufficient brake force

A brake system that is capable of providing sufficient brake power, giving the user a feeling of control and safety.

2. Axiomatic design

Step 2: Top-level FRs and DPs

- Based on the 440 customer statements collected and the expertise of the General Manager, FRs and DPs were formulated

Table 1. Top-level parameters for electric bicycle design.

Index	FR	DP
1	Provide basic structure	Frame assembly
2	Allow movement	Wheelset
3	Drive electrically	Electric drive assembly
4	Drive mechanically	Mechanical drive assembly
5	Interact with user	Human interaction components
6	Have optional functions	Flexible accessories packages
7	Have sufficient brake force	Quality hydraulic disc brakes
8	Have lighting	Lighting package, frame mount

$$\begin{Bmatrix} FR_1 \\ FR_2 \\ FR_3 \\ FR_4 \\ FR_5 \\ FR_6 \\ FR_7 \\ FR_8 \end{Bmatrix} = \begin{bmatrix} X & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ X & X & 0 & 0 & 0 & 0 & 0 & 0 \\ X & X & X & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & X & X & 0 & 0 & 0 & 0 \\ 0 & X & 0 & 0 & X & 0 & 0 & 0 \\ X & 0 & 0 & 0 & 0 & X & 0 & 0 \\ X & 0 & X & 0 & 0 & 0 & X & 0 \\ X & 0 & 0 & 0 & 0 & 0 & 0 & X \end{bmatrix} \begin{Bmatrix} DP_1 \\ DP_2 \\ DP_3 \\ DP_4 \\ DP_5 \\ DP_6 \\ DP_7 \\ DP_8 \end{Bmatrix}$$

2. Axiomatic design

Step 3: Decomposition of FR and DP

Index	FR	DP
11	High stiffness	Rigid aluminium construction
12	Good-looking weld seams	Quality welding supplier
13	Frame shock absorption	Suspension front fork
14	Unique design	Double top-tube
15	Easy to clean	Wet paint

Index	FR	DP
21	High tyre shock absorption	Large diameter tyres
22	Low rolling resistance	Low friction profile

Index	FR	DP
41	Gearshift	Derailleur gears
42	Compatibility with electric drive	Single speed crankset

Index	FR	DP
51	Saddle shock absorption	Comfortable saddle
52	Steering control	Quality grips

Index	FR	DP
31	Propulsion	Motor-/gearbox unit with integrated controller
32	Interaction with electrical drive	High value display
33	Speed metering	Speed sensor, spoke magnet based method
34	Innovative charging	Inductive charging system
35	Control of the electric drive	Controller software
36	Assist during walking	Walking assist
37	Good gearshift performance	Gear sensor
38	Sufficient range	Lithium-ion battery

Index	FR	DP
321	Good usability	Graphical user interface (GUI)
322	High contrast	High contrast screen
323	Sufficient functionality	3 drive modes

Index	FR	DP
361	Software function	5 km/h limiter software module
362	User interface	On/off button

2. Axiomatic design

Step 4: Design matrix

- Mapping in between the functional domain and the physical domain, resulting in the **design matrix**

		DPs																															
		1				2				3				4				5				6				7				8			
		1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8
FRs	1	x																															
	2		x																														
	3			x																													
	4				x																												
	5					x																											
	6						x																										
	7							x																									
	8								x																								
		1				2				3				4				5				6				7				8			
		1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8

Decoupling

		DPs																															
		1				2				3				4				5				6				7				8			
		1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8
FRs	1	x																															
	2		x																														
	3			x																													
	4				x																												
	5					x																											
	6						x																										
	7							x																									
	8								x																								

Figure 16. Design matrix prior to decoupling.

2. Axiomatic design

Step 5: Design process

- The presented design was not achieved at once, it is the result of a process of **re-design cycles**
- In the first step, FR and DP hierarchies had to be made consistent, otherwise there would have been insufficient DPs or a redundant design. This was achieved through **combination, decomposition and zigzagging in between FRs and DPs**.
- Improving design quality by adding or removing dependencies in between FRs and DPs as well as resolving coupling issues



2. Axiomatic design

Step 5: RESULTS

- Axiomatic Design helps to analyse the capabilities and visualizes the dependencies of the single components
- Many mistakes can be shown at an early stadium, so time can be saved without getting these results too late in a more advanced stage
- **Limitation:** the workload that comes with the quantitative assessment of the Information Axiom



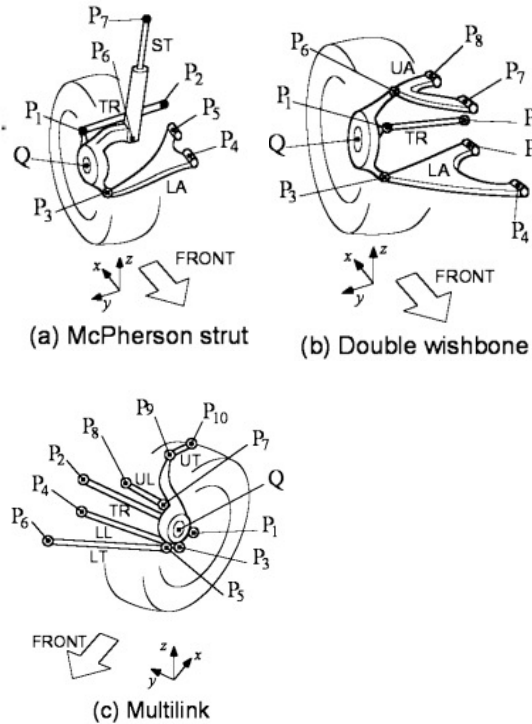
2. Axiomatic design

Case study - Automotive Suspension Systems

FR_1 = maintain desired toe angle change.
 FR_2 = maintain desired camber angle change.
 FR_3 = maintain desired wheelbase change.
 FR_4 = maintain desired kingpin offset.
 FR_5 = maintain desired caster angle.

DP_1 = toe control link (tie rod).
 DP_2 = front view of suspension layout.
 DP_3 = side view of suspension layout.
 DP_4 = front view of kingpin axis.
 DP_5 = side view of kingpin axis.

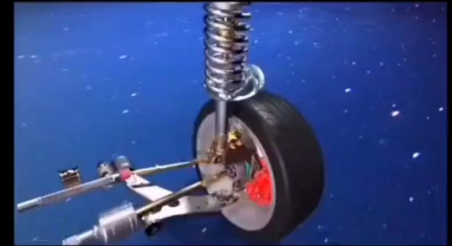
Axiomatic Design of Automotive Suspension Systems, Sangwoo Bae



Double wishbone Suspension



MacPherson Suspension

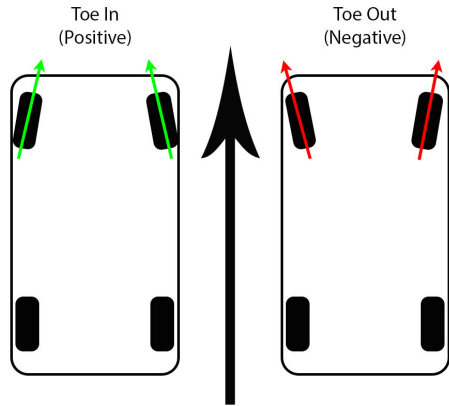


Multi-link Independent



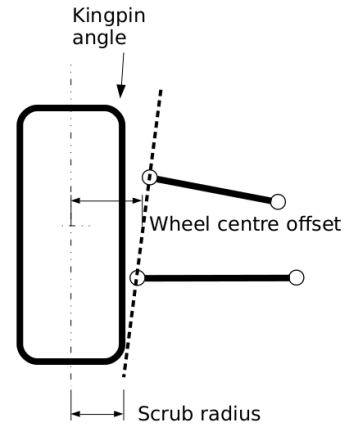
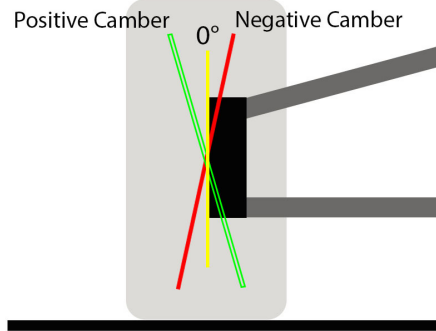
2. Axiomatic design

Toe Angle



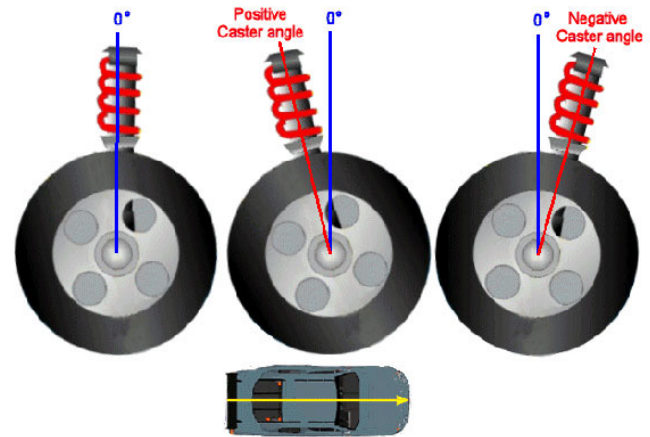
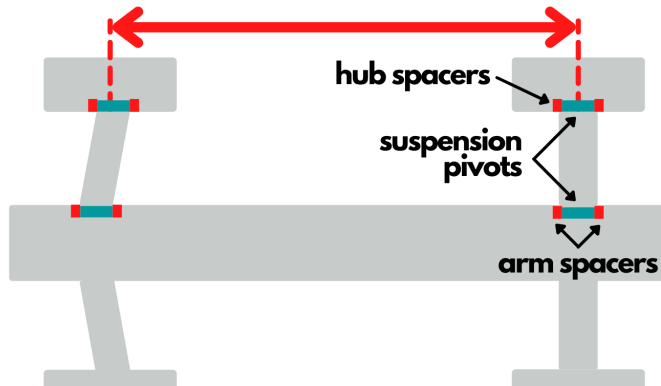
top view

Camber Angle



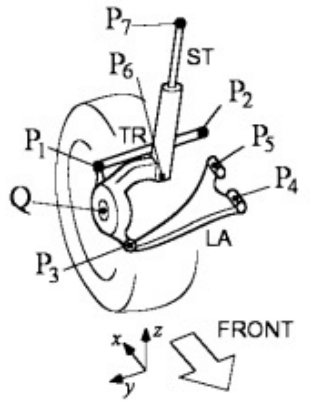
wheelbase

so dialed



2. Axiomatic design

McPherson strut



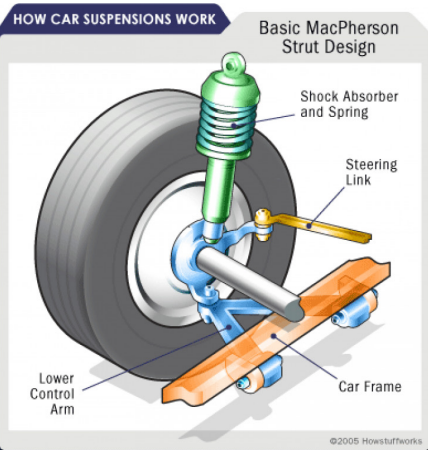
(a) McPherson strut

- This type of suspension consists of an arm (LA), a strut (ST) and a link (TR)
- It has 7 hardpoints (P1 - P7) which have three coordinates respectively, resulting 21 DPs

Table 1: Effect of DPs on FRs for the McPherson strut.

H/P	DPs	FR ₁	FR ₂	FR ₃	FR ₄	FR ₅
P ₁	DP ₃	-X ✓	+0	+⊗	+0	-0
P ₂	DP ₆	+X ✓	-0	-⊗	-0	+0
	DP ₇	+0	+0	-0	-0	+X ✓
P ₃	DP ₈	+0	+0	+0	+X ✓	+0
	DP ₉	+X	+X	-⊗	+⊗ ✓	-0
P ₄	DP ₁₂	-X	-X	+X ✓	+0	+0
P ₅	DP ₁₅	-0	+0	-X ✓	-0	-0
P ₆	DP ₁₇	+⊗	-X ✓	-0	-0	+0
P ₇	DP ₁₉	+0	-0	+0	+0	-X ✓

(Strong,
passable,
weak)



a cross-coupled system

$$DP'_1 = DP_3 \text{ and } DP_6$$

$$DP'_2 = DP_{17}$$

$$DP'_3 = DP_{12} \text{ and } DP_{15}$$

$$DP'_4 = DP_8 \text{ and } DP_9$$

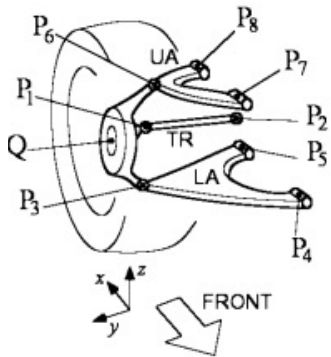
$$DP'_5 = DP_7 \text{ and } DP_{19}$$

$$\begin{Bmatrix} FR_5 \\ FR_4 \\ FR_3 \\ FR_2 \\ FR_1 \end{Bmatrix} = \begin{bmatrix} \times & 0 & 0 & 0 & 0 \\ 0 & \times & 0 & 0 & 0 \\ 0 & 0 & \times & 0 & \otimes \\ 0 & \times & \times & \times & 0 \\ 0 & \times & \times & 0 & \times \end{bmatrix} \begin{Bmatrix} DP'_5 \\ DP'_4 \\ DP'_3 \\ DP'_2 \\ DP'_1 \end{Bmatrix}$$

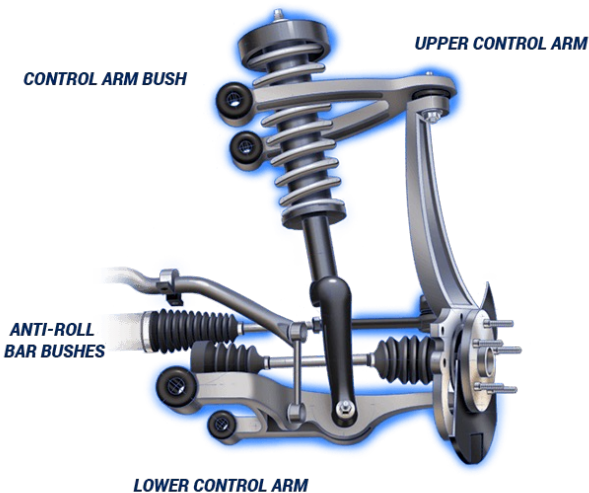
2. Axiomatic design

Double wishbone

- The double wishbone suspension consists of two arms (UA, LA) and a link (TR)
- With 8 hardpoints (P1 - P_a), there are a total of 24 DPs



(b) Double wishbone



a cross-coupled system

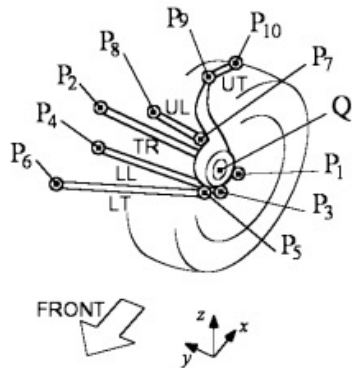
H/P	DPs	FR ₁	FR ₂	FR ₃	FR ₄	FR ₅
P ₁	DP ₃	-X ✓	-0	-X	-0	-0
P ₂	DP ₆	+X ✓	+0	+X	+0	+0
P ₃	DP ₇	+0	+⊗	+0	+0	+X ✓
	DP ₈	-0	-0	-0	-X ✓	-0
P ₄	DP ₁₂	-⊗	-⊗	-X ✓	-0	+0
P ₅	DP ₁₅	-⊗	-⊗	+X ✓	-0	-0
	DP ₁₆	-0	+0	+0	-0	-X ✓
P ₆	DP ₁₇	-0	+0	-0	+X ✓	+0
	DP ₁₈	+⊗	-X ✓	+0	+⊗	-0
P ₇	DP ₂₁	-0	+⊗ ✓	-0	-0	-0

$$\begin{Bmatrix} \text{FR}_5 \\ \text{FR}_4 \\ \text{FR}_3 \\ \text{FR}_2 \\ \text{FR}_1 \end{Bmatrix} = \begin{bmatrix} \times & 0 & 0 & 0 & 0 \\ 0 & \times & 0 & 0 & 0 \\ 0 & 0 & \times & 0 & \otimes \\ 0 & 0 & \times & \times & 0 \\ 0 & 0 & \times & \times & \times \end{bmatrix} \begin{Bmatrix} \text{DP}'_5 \\ \text{DP}'_4 \\ \text{DP}'_3 \\ \text{DP}'_2 \\ \text{DP}'_1 \end{Bmatrix}$$

$$\begin{aligned} \text{DP}'_1 &= \text{DP}_3 \text{ and } \text{DP}_6 \\ \text{DP}'_2 &= \text{DP}_{18} \text{ and } \text{DP}_{21} \\ \text{DP}'_3 &= \text{DP}_{12} \text{ and } \text{DP}_{15} \\ \text{DP}'_4 &= \text{DP}_8 \text{ and } \text{DP}_{17} \\ \text{DP}'_5 &= \text{DP}_7 \text{ and } \text{DP}_{16} \end{aligned}$$

2. Axiomatic design

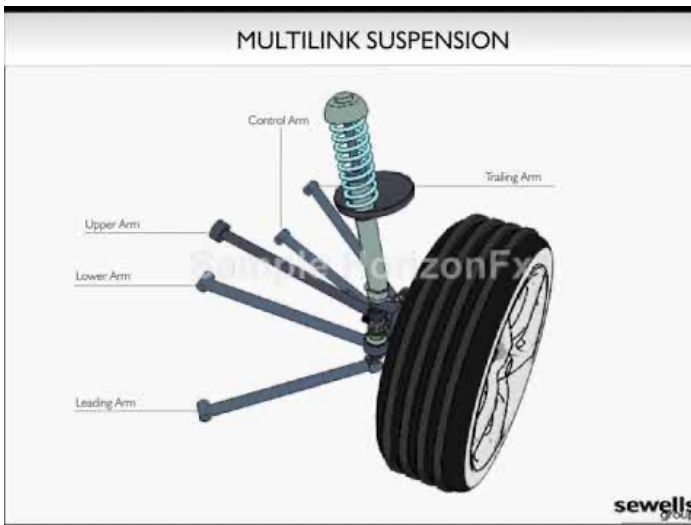
Multilink



(c) Multilink

- The multilink suspension consists of five links (UL, UT, LL, LT, TR)
- With 10 hardpoints (P1 – P10), there are 30 DPs
- P4, P9 and P10 are not selected as the dominant DPs, and therefore they are removed for limited space

H/P	DPs	FR ₁	FR ₂	FR ₃	FR ₄	FR ₅
P ₁	DP ₃	+X ✓	+0	+⊙	+0	-0
P ₂	DP ₆	-X ✓	-0	-⊙	-0	+0
P ₃	DP ₇	-0	+0	+0	+X	+X ✓
	DP ₁₃	-0	+0	-0	-X ✓	+0
P ₅	DP ₁₄	+0	-0	-0	+X ✓	-0
	DP ₁₅	-0	+0	-X ✓	+0	-0
P ₆	DP ₁₈	+0	-0	+X ✓	+0	-0
	DP ₁₉	-0	-0	-0	+0	-X ✓
P ₇	DP ₂₁	-0	-X ✓	-0	-0	+0
P ₈	DP ₂₄	+0	+X ✓	+0	-0	+0



sewells

Fully decoupled system

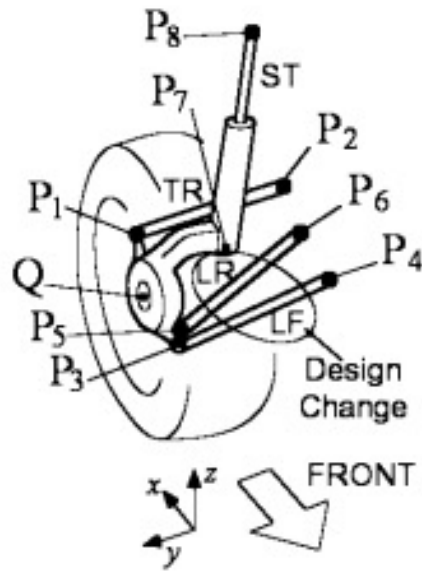
$$\begin{aligned}
 DP'_1 &= DP_3 \text{ and } DP_6 \\
 DP'_2 &= DP_{21} \text{ and } DP_{24} \\
 DP'_3 &= DP_{15} \text{ and } DP_{18} \\
 DP'_4 &= DP_{13} \text{ and } DP_{14} \\
 DP'_5 &= DP_7 \text{ and } DP_{19}
 \end{aligned}$$

$$\begin{Bmatrix} FR_5 \\ FR_4 \\ FR_2 \\ FR_1 \\ FR_3 \end{Bmatrix} = \begin{bmatrix} \times & 0 & 0 & 0 & 0 \\ \times & \times & 0 & 0 & 0 \\ 0 & 0 & \times & 0 & 0 \\ 0 & 0 & 0 & \times & 0 \\ 0 & 0 & \times & \times & \times \end{bmatrix} \begin{Bmatrix} DP'_5 \\ DP'_4 \\ DP'_2 \\ DP'_1 \\ DP'_3 \end{Bmatrix}$$

2. Axiomatic design

Design change for the McPherson strut

the LA is replaced with two equivalent links (LF, LR) to convert it to a **decoupled design**



$$\begin{Bmatrix} FR_5 \\ FR_4 \\ FR_2 \\ FR_1 \\ FR_3 \end{Bmatrix} = \begin{bmatrix} \times & 0 & 0 & 0 & 0 \\ \times & \times & 0 & 0 & 0 \\ 0 & 0 & \times & 0 & 0 \\ 0 & 0 & 0 & \times & 0 \\ 0 & 0 & 0 & \times & \times \end{bmatrix} \begin{Bmatrix} DP'_5 \\ DP'_4 \\ DP'_2 \\ DP'_1 \\ DP'_3 \end{Bmatrix}$$

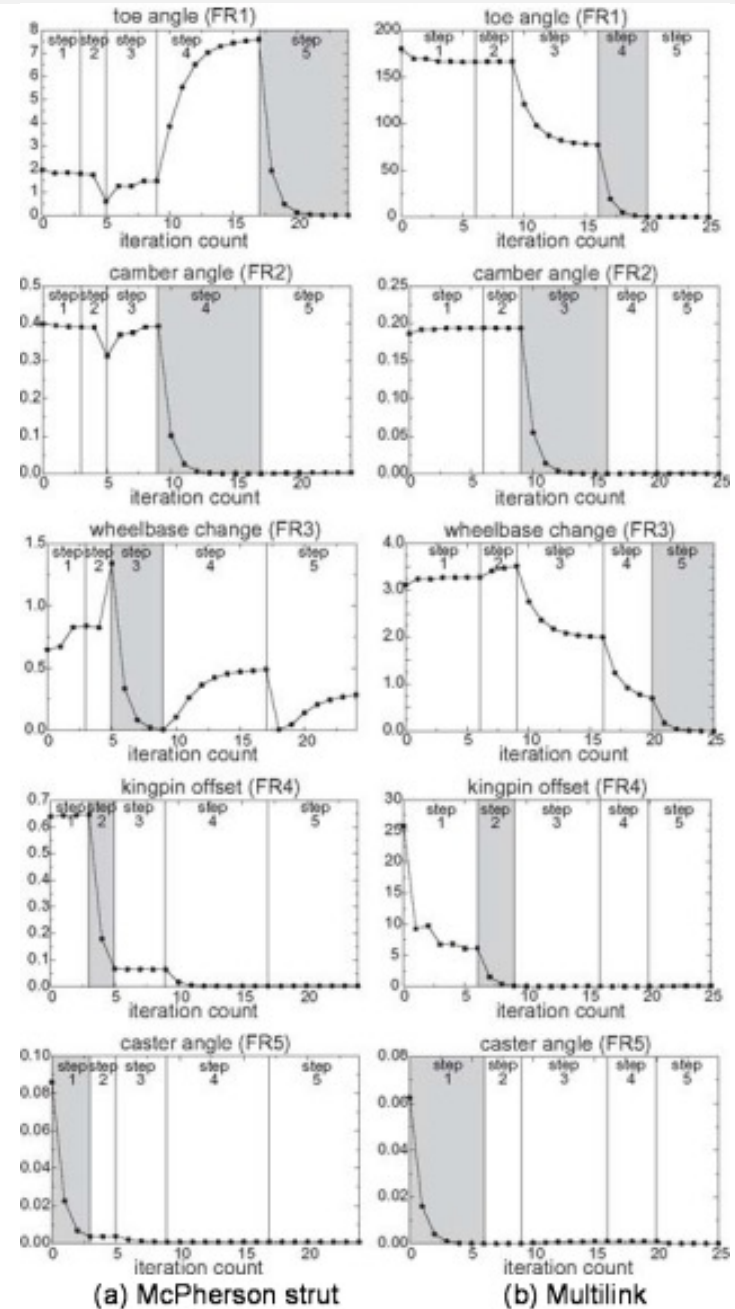
Design change for the McPherson strut.

decoupled system

2. Axiomatic design

SEQUENTIAL DESIGN PROCEDURE

- **McPherson strut and double wishbone:** FR5, FR4, FR3, FR2 and FR1
- **Multilink and design change:** detailed design should start with the caster (FR5), followed by the kingpin offset (FR4), camber (FR2), toe (FR1) and wheelbase change (FR3).



2. Axiomatic design

Case study - Lift stacker

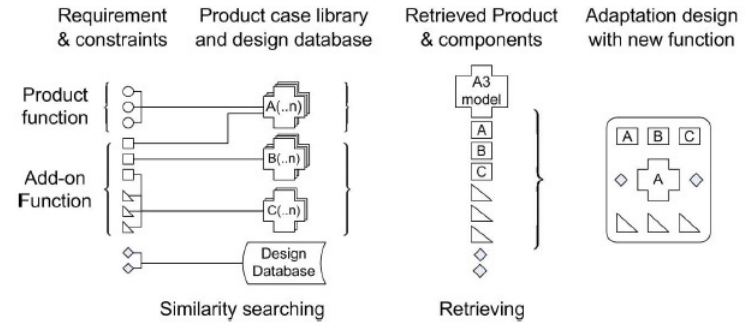


Figure 2: Design Methodology

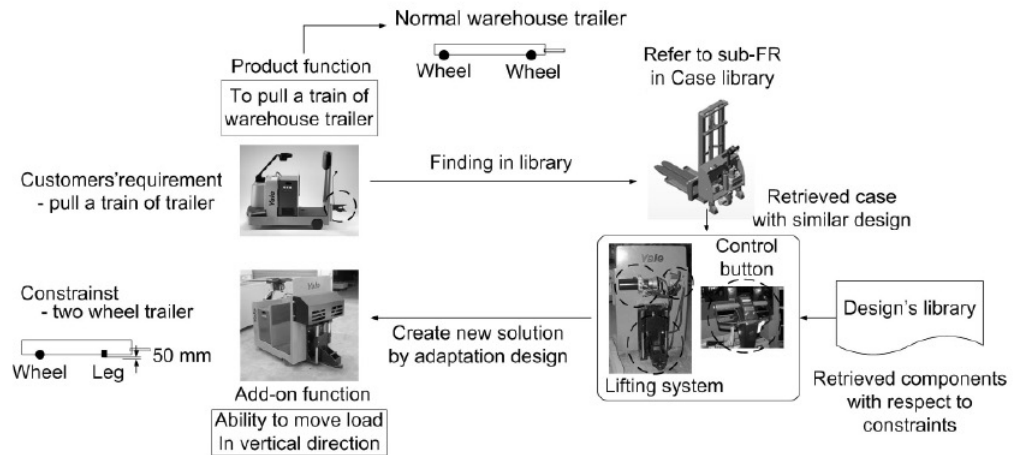


Figure 3: An example of add-on functions

2. Axiomatic design

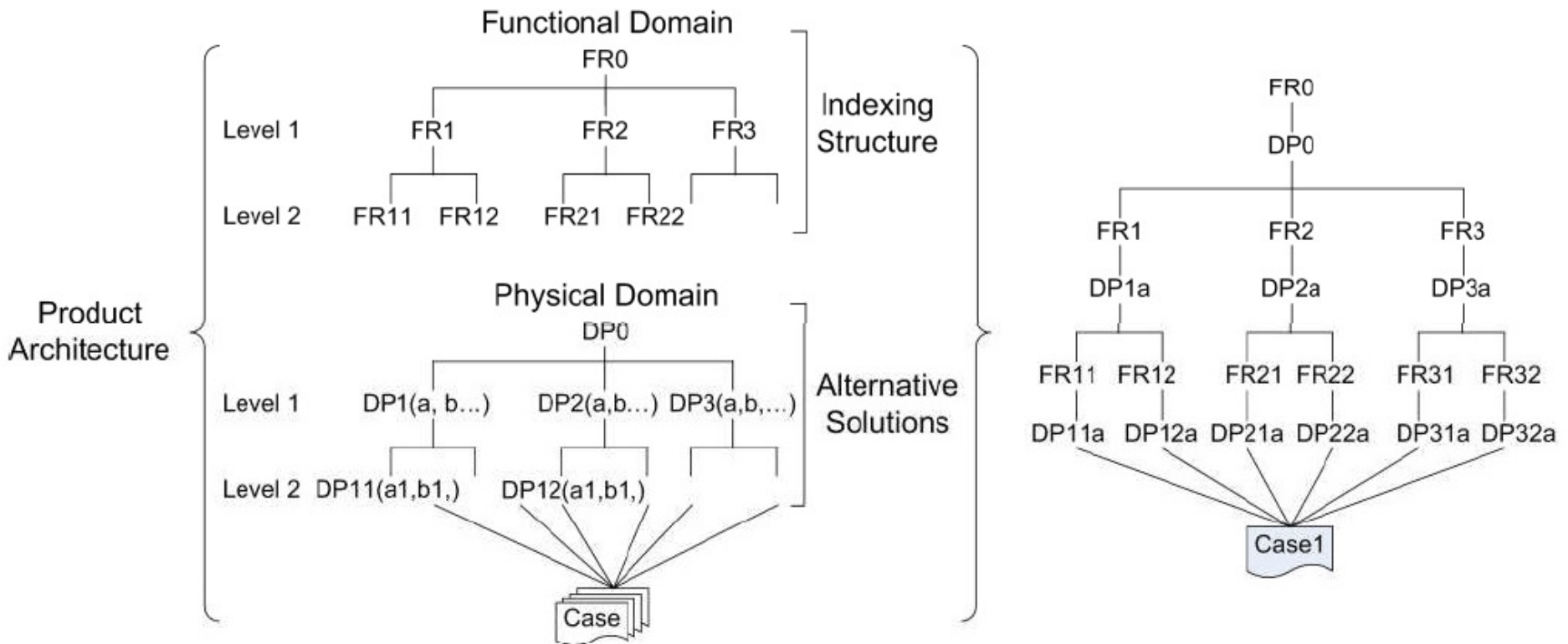
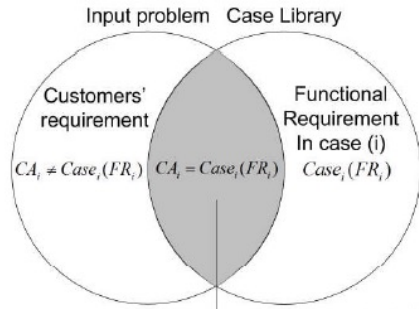


Figure 5: An index structure in case library

2. Axiomatic design



Find the most similar case
in the set

$$Sim(CS_i, Case(DP_i)) = \sum_1^n 1 - \frac{(CS_i - Case_i(DP_i))}{system\ range(DP_i)}$$

